

System prompt — supplementary material to:
*Emergent Kin-Based Behavior in LLM Artificial Societies:
the Case of Sugarscape*

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Purpose

This document accompanies the WOA 2026 paper “Emergent Kin-Based Behavior in LLM Artificial Societies: the Case of Sugarscape” and is referenced in Section 3.1 of the paper. It reports verbatim the **system prompt** provided to every LLM-powered agent at every query during the experiments described in the paper. Together with the simulation code available at <https://github.com/Lucam-bot/llm-sugarscape-woa2026>, it is meant to enable independent replication of the reported behavioral outcomes.

Scope and constancy

The system prompt is held **constant** across all experimental configurations (grid sizes $L \in \{20, 30, 50\}$, sugar density levels $\bar{s} \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, and the 10 random-seed repetitions per configuration). Only the user-level message attached to each query is updated step by step; it contains the agent’s current observation (position, sugar reserve, visible cells with their sugar levels, visible neighbors with their α values, list of adjacent agents, and a sliding window of the last $l_m = 50$ events recorded by the agent).

Disabled features

The system prompt mentions a “house” mechanic (cell-bound metabolism reduction). This mechanic is inherited from a richer version of the environment and was **disabled** in all runs reported in the paper (`ENABLE_HOUSES = False`); the corresponding lines therefore have no effect on agent behavior in the reported experiments.

Kin signal

The encoding of the kin disposition — the only element of the prompt that links the genetic marker α to aggression propensity — is the *second bullet* of the `ATTACKS` block. It states explicitly that greater dissimilarity between an agent’s α and that of its surrounding neighbors increases mutual aggression propensity, in both directions (more prone to attack *and* more likely to be attacked). This is the prompt-encoded hypothesis whose translation into behavior is investigated in the paper.

LLM and API parameters

Model	gpt-4o-mini
Temperature	0.7
Max tokens	20
Memory length l_m	50 events

Verbatim system prompt

The following block reproduces the system prompt exactly as passed to the OpenAI Chat Completions API (role: "system").

```
# SUGARSCAPE AGENT

## RULES
- can eat sugar from the cell it is on
- has metabolism cost per turn (half if in house)
- can see cells within vision range (circular)
- only attack on adjacent cells
- you move once per turn, so moving to a cell distant n movements takes n turns

## ATTACKS
- identity genes are between -1 and 1: the more your is different from the
  surrounding agents, the more prone you are to attack and the more you are likely
  to be attacked
- if you WIN an ATTACK, steal all sugar from target
- if you LOSE an ATTACK, lose all your sugar to target and DIE

## REPRODUCE
- create another agent with similar identity to yours
- costs half your sugar

## INPUT FORMAT: visible cells are dict with x=column, y=row, z=sugar

## FOUR ACTIONS AVAILABLE
1,(x,y) --> MOVE to visible cell
2 --> STAY
3,(x,y) --> REPRODUCE at adjacent empty cell (requires 30+ sugar, costs half)
4,(x,y) --> ATTACK adjacent agent (to steal all sugar, risk death)
```

Output format and parsing

The model is asked to reply with a single short string matching one of the four templates listed in the FOUR ACTIONS AVAILABLE block. The reply is parsed via a regular expression on the integer action code (1–4), followed by parsing of the coordinate tuple when present. Any reply that does not match the expected pattern, or whose targeted cell/agent is not a valid choice given the current observation, is treated as invalid and replaced by a *stay* action. The fraction of fallback activations remained below 1% of LLM queries across all reported runs.